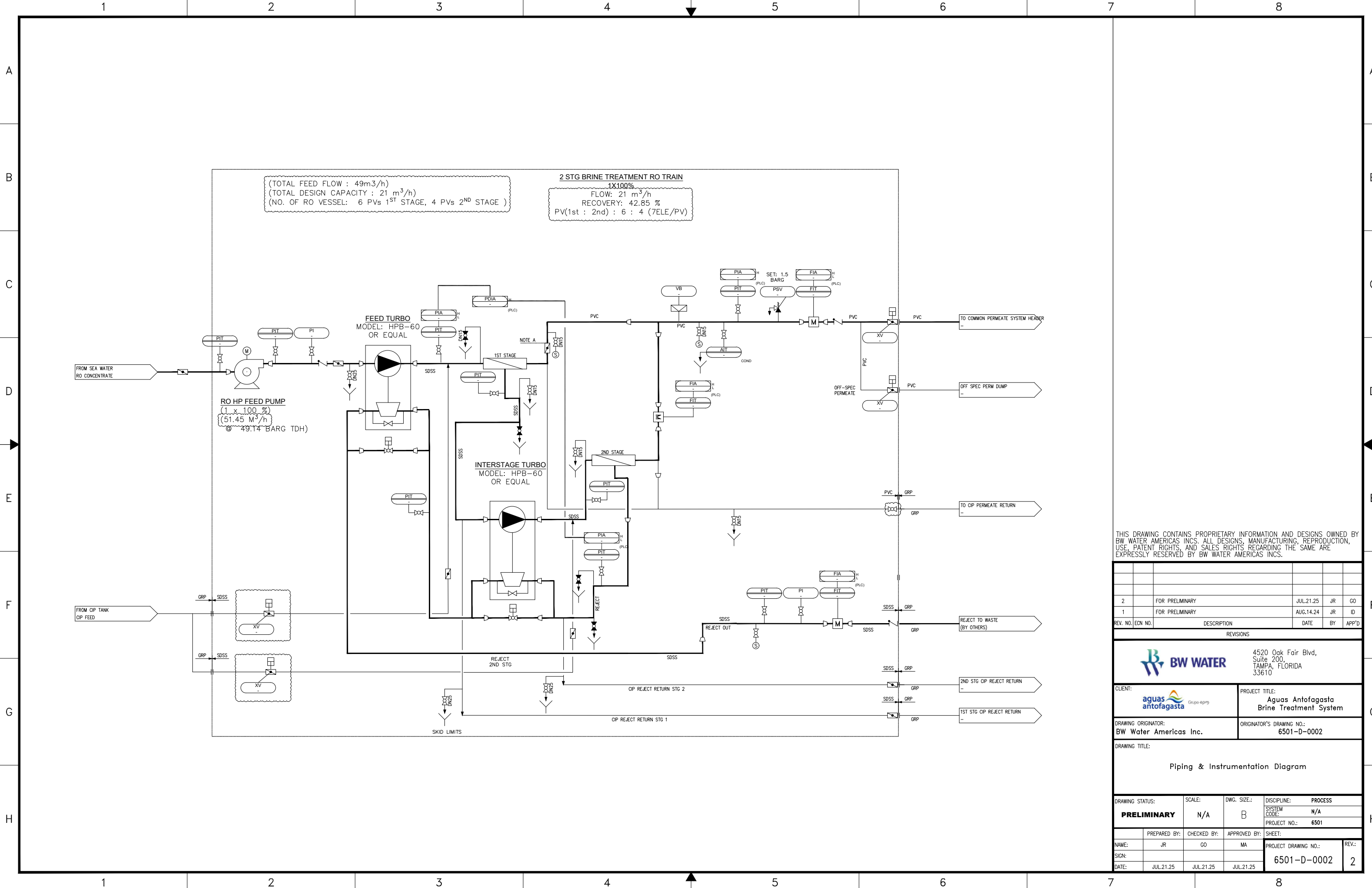




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2		FOR PRELIMINARY	JUL.21.25	JR	GO
1		FOR PRELIMINARY	AUG.14.24	JR	ID
REV. NO.	ECN NO.	DESCRIPTION	DATE	BY	APP'D

REVISIONS

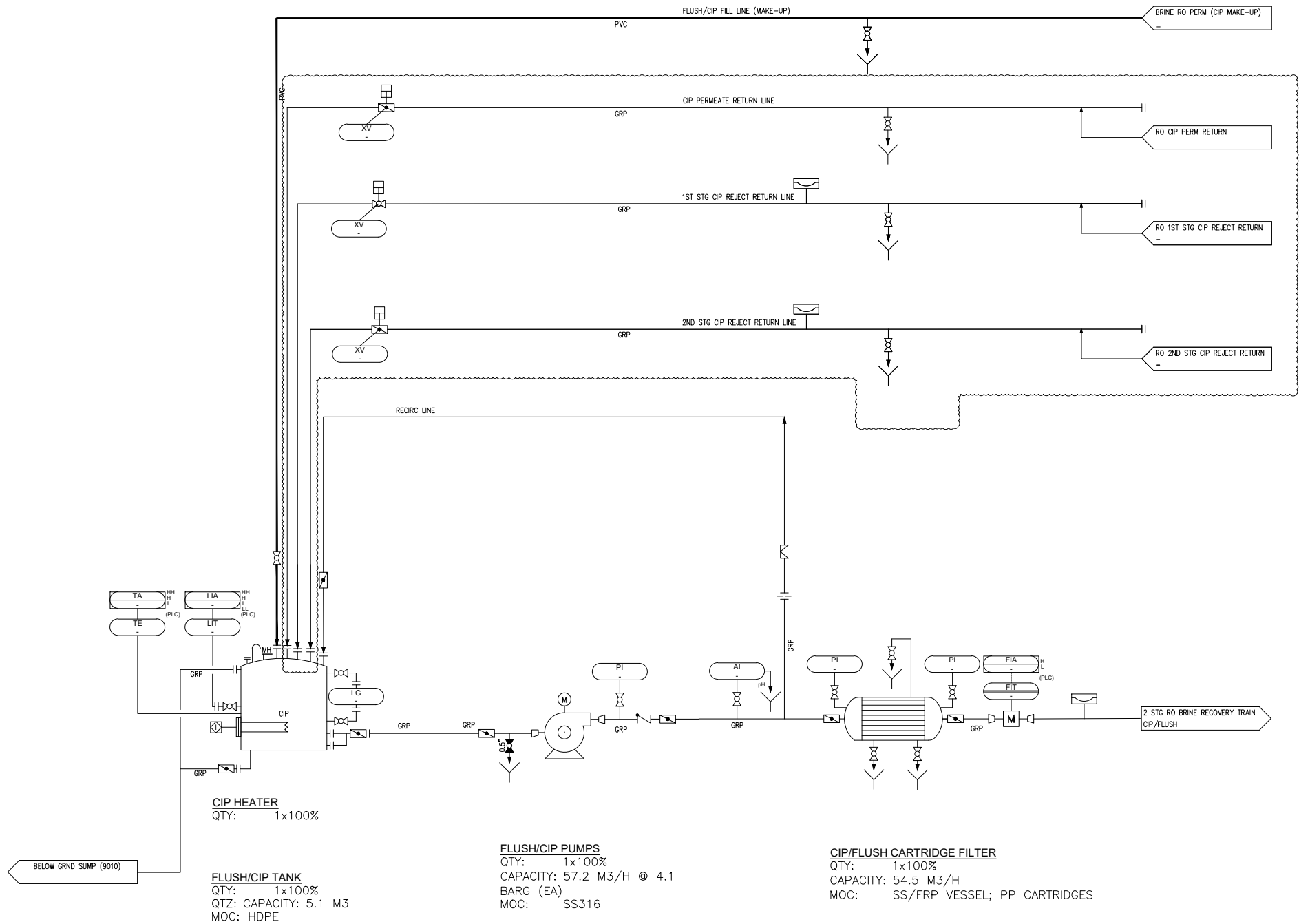


4520 Oak Fair Blvd,
Suite 200,
TAMPA, FLORIDA
33610

CLIENT: 	PROJECT TITLE: Aguas Antofagasta Brine Treatment System
DRAWING ORIGINATOR: BW Water Americas Inc.	ORIGINATOR'S DRAWING NO.: 6501-D-0002

Piping & Instrumentation Diagram

DRAWING STATUS: PRELIMINARY	SCALE: N/A	DWG. SIZE.: B	DISCIPLINE: PROCESS	SYSTEM CODE: N/A	PROJECT NO.: 6501
NAME: SIGN:	PREPARED BY: JR	CHECKED BY: GO	APPROVED BY: MA	SHEET: PROJECT DRAWING NO.: 6501-D-0002	REV.: 2
DATE:	JUL.21.25	JUL.21.25	JUL.21.25		



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REV. NO.	ECN NO.	DESCRIPTION	DATE	BY	APP'D
2		FOR PRELIMINARY	JUL.21.25	JR	GO
1		FOR PRELIMINARY	AUG.14.24	JR	ID

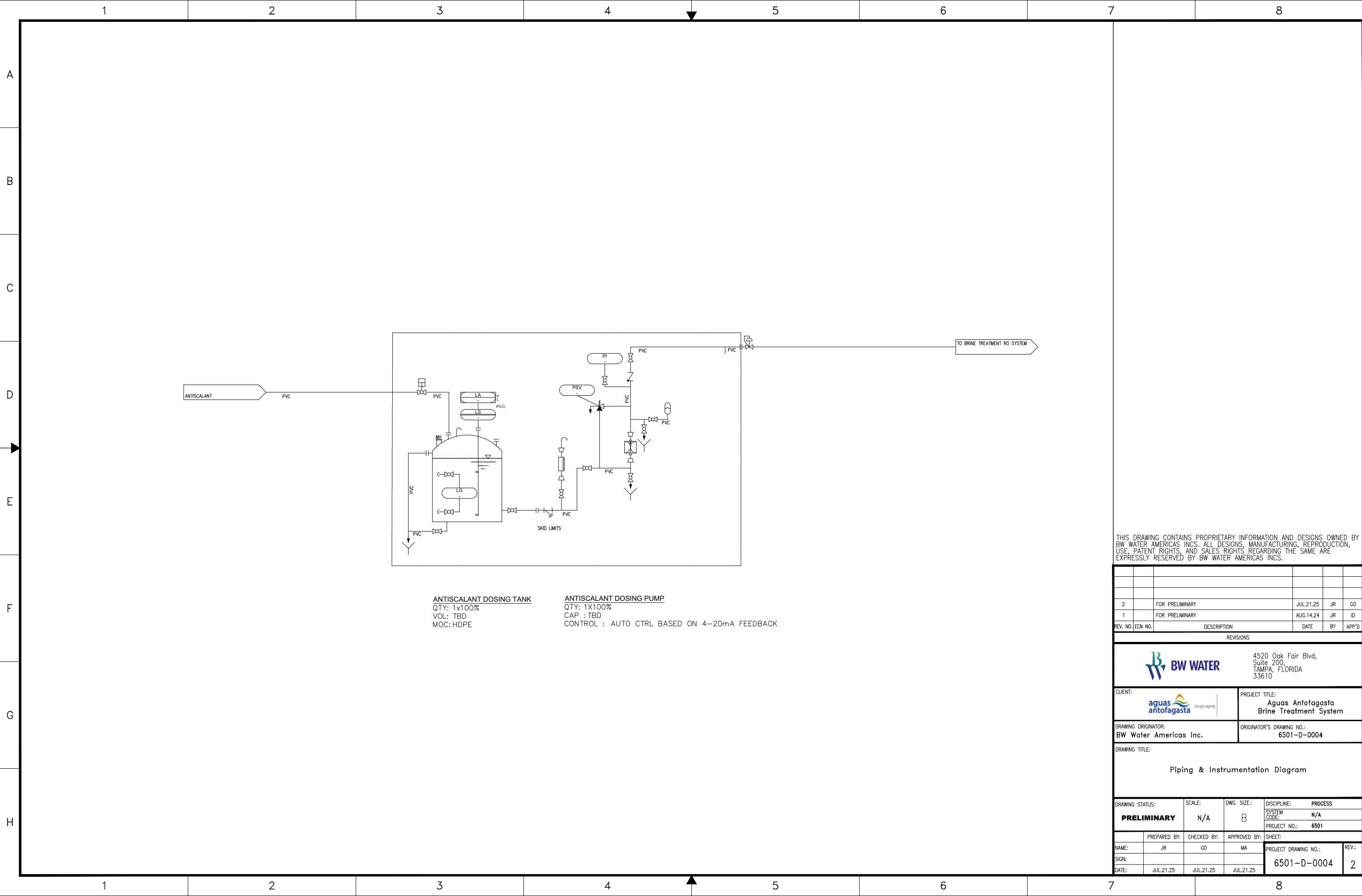


4520 Oak Fair Blvd,
Suite 200,
TAMPA, FLORIDA
33610

CLIENT: 	PROJECT TITLE: Aguas Antofagasta Brine Treatment System
DRAWING ORIGINATOR: BW Water Americas Inc.	ORIGINATOR'S DRAWING NO.: 6501-D-0003

DRAWING TITLE: Piping & Instrumentation Diagram				
DRAWING STATUS: PRELIMINARY	SCALE: N/A	DWG. SIZE: B	DISCIPLINE: PROCESS	SYSTEM CODE: N/A
			PROJECT NO.: 6501	

PREPARED BY: JR	CHECKED BY: GO	APPROVED BY: MA	SHEET: PROJECT DRAWING NO.: 6501-D-0003	REV.: 2
DATE: JUL.21.25	DATE: JUL.21.25	DATE: JUL.21.25		



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2		FOR PRELIMINARY	JUL.21.25	JR	GO
1		FOR PRELIMINARY	AUG.14.24	JR	ID
REV. NO.	ECN NO.	DESCRIPTION	DATE	BY	APP'D

REVISIONS					
			4520 Oak Fair Blvd, Suite 200, TAMPA, FLORIDA 33610		

CLIENT:		PROJECT TITLE:	Aguas Antofagasta Brine Treatment System		
DRAWING ORIGINATOR:	BW Water Americas Inc.	ORIGINATOR'S DRAWING NO.:	6501-D-0004		

DRAWING TITLE:					
Piping & Instrumentation Diagram					

DRAWING STATUS:	SCALE:	DWG. SIZE.:	DISCIPLINE:	PROCESS
PRELIMINARY	N/A	B	SYSTEM CODE:	N/A
			PROJECT NO.:	6501
	PREPARED BY:	CHECKED BY:	APPROVED BY:	SHEET:
NAME:	JR	GO	MA	PROJECT DRAWING NO.:
SIGN:				6501-D-0004
DATE:	JUL.21.25	JUL.21.25	JUL.21.25	REV.: 2

Technical Proposal

7. RO MEMBRANE PROJECTIONS

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Modelaciones con las membranas seleccionadas para membranas nuevas y cinco años de uso”)

The following RO membrane projections are included:

Conditions: TDS 43,000 ppm | Temperature: 19°C

- Year 0
- Year 1
- Year 5

Conditions: TDS 43,000 ppm | Temperature: 24°C

- Year 0
- Year 1
- Year 5

Conditions: TDS 53,000 ppm | Temperature: 19°C

- Year 0
- Year 1
- Year 5

Conditions: TDS 53,000 ppm | Temperature: 24°C

- Year 0
- Year 1
- Year 5

Please also refer to Section 3.2 for notes related to the performance-limited warranty provided by the UHPRO membrane manufacturer.

Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 43K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 0
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 43,186.82 mg/L
Feed osmotic pressure: 30.4 bar
Concentrate osmotic pressure: 53.15 bar

Feed pressure: 48.57 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 l/mh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 43,186.82 mg/L
Feed osmotic pressure: 30.4 bar
Concentrate osmotic pressure: 53.15 bar
Pump efficiency: 80 %

Temperature: 19 °C
Average NDP: 10.51 bar
Specific energy: 4.58 kWh/m³
Feed pressure: 48.57 bar
Permeate TDS: 193.73 mg/L
Fouling factor: 1

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.07	35.93	48.57	47.48	1.09	0	1	0	8.81	166.96
Stage 2	4	7	35.93	7.95	27.98	61.48	60.13	1.35	14	1	0	8.04	237.73

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	13,445.30	13,445.30	18,314.73	23,494.76	23,494.76	59.85	85.26	69.46
Potassium	473.70	473.70	645.13	827.44	827.44	2.48	3.54	2.88
Magnesium	1,461.94	1,461.94	1,993.25	2,559.08	2,559.08	1.41	2.02	1.64
Calcium	505.56	505.56	689.30	884.97	884.97	0.48	0.69	0.56
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	23,704.79	23,704.79	32,292.67	41,429.29	41,429.29	97.79	139.31	113.50
Sulfate	3,362.28	3,362.28	4,584.93	5,887.21	5,887.21	1.37	1.96	1.59
Nitrate	0.16	0.16	0.22	0.28	0.28	0.00	0.01	0.01
Carbonate	2.11	2.11	2.88	3.69	3.69	0.00	0.00	0.00
Bicarbonate	221.40	221.40	301.13	385.79	385.79	2.24	3.18	2.60
Boron	7.48	7.48	9.72	11.98	11.98	1.32	1.76	1.49
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.86	3.67	3.67	0.01	0.01	0.01
CO2	6.36	6.36	6.36	6.36	6.36	6.36	6.36	6.36
TDS	43,186.82	43,186.82	58,836.83	75,488.18	75,488.18	166.96	237.73	193.73
pH	7.60	7.60	7.72	7.82	7.82	5.78	5.93	5.84

Within Vessels - Pass1

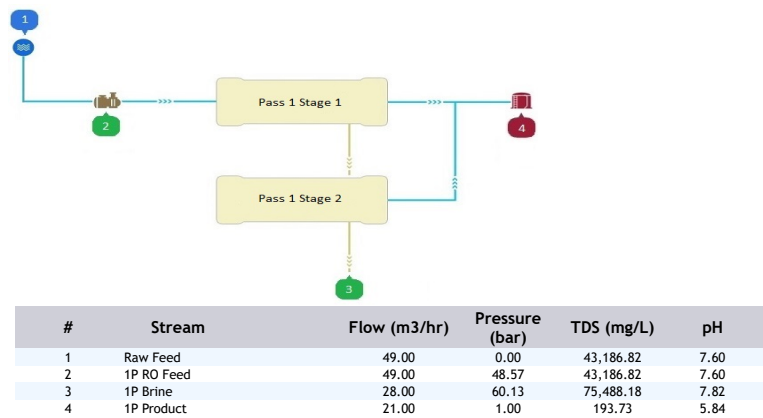
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.53	14.93	6.45	0.20	12.96	1.10	43,180.01	87.82
LG SW 400 R G2 UHP	2	7.64	0.43	12.29	5.68	0.18	11.01	1.09	46,153.15	112.12
LG SW 400 R G2 UHP	3	7.21	0.35	10.04	4.92	0.16	9.28	1.07	48,925.69	143.09
LG SW 400 R G2 UHP	4	6.85	0.29	8.16	4.21	0.15	7.77	1.06	51,450.57	182.33
LG SW 400 R G2 UHP	5	6.56	0.23	6.61	3.56	0.14	6.47	1.05	53,702.20	231.73
LG SW 400 R G2 UHP	6	6.33	0.19	5.35	2.98	0.13	5.37	1.04	55,674.42	293.51
LG SW 400 R G2 UHP	7	6.14	0.15	4.32	2.48	0.13	4.44	1.04	57,376.52	370.23
Stage 2										
LG SW 400 R G2 UHP	1	8.98	0.48	13.62	5.35	0.23	14.32	1.08	58,828.39	128.61
LG SW 400 R G2 UHP	2	8.50	0.39	11.14	4.62	0.21	12.26	1.07	62,147.89	163.53
LG SW 400 R G2 UHP	3	8.11	0.32	9.09	3.96	0.20	10.46	1.06	65,153.71	207.06
LG SW 400 R G2 UHP	4	7.79	0.26	7.41	3.36	0.19	8.89	1.05	67,829.94	260.98
LG SW 400 R G2 UHP	5	7.53	0.21	6.05	2.84	0.18	7.55	1.04	70,179.18	327.14
LG SW 400 R G2 UHP	6	7.31	0.17	4.94	2.39	0.17	6.39	1.04	72,218.91	407.69
LG SW 400 R G2 UHP	7	7.14	0.14	4.05	2.00	0.16	5.41	1.03	73,975.17	504.98

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	0.73	1.64
CaSO4	28.5 %	59.07 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.86	0.11

Warnings - Pass1

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Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 43K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 1
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 43,186.82 mg/L
Feed osmotic pressure: 30.4 bar
Concentrate osmotic pressure: 53.14 bar

Feed pressure: 49.17 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 43,186.82 mg/L
Feed osmotic pressure: 30.4 bar
Concentrate osmotic pressure: 53.14 bar
Pump efficiency: 80 %

Temperature: 19 °C
Average NDP: 11.13 bar
Specific energy: 4.63 kWh/m³
Feed pressure: 49.17 bar
Permeate TDS: 207.01 mg/L
Fouling factor: 0.93

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.1	35.89	49.17	48.08	1.09	0	1	0	8.84	177.98
Stage 2	4	7	35.89	7.92	27.98	62.08	60.73	1.35	14	1	0	8.01	255.08

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	13,445.30	13,445.30	18,330.83	23,491.99	23,491.99	63.80	91.49	74.23
Potassium	473.70	473.70	645.69	827.32	827.32	2.65	3.80	3.08
Magnesium	1,461.94	1,461.94	1,995.13	2,559.08	2,559.08	1.51	2.16	1.76
Calcium	505.56	505.56	689.95	884.97	884.97	0.52	0.74	0.60
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	23,704.79	23,704.79	32,321.26	41,424.91	41,424.91	104.25	149.50	121.29
Sulfate	3,362.28	3,362.28	4,589.30	5,887.34	5,887.34	1.46	2.10	1.70
Nitrate	0.16	0.16	0.22	0.28	0.28	0.01	0.01	0.01
Carbonate	2.11	2.11	2.88	3.69	3.69	0.00	0.00	0.00
Bicarbonate	221.40	221.40	301.36	385.67	385.67	2.39	3.42	2.77
Boron	7.48	7.48	9.70	11.92	11.92	1.40	1.86	1.57
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.86	3.67	3.67	0.01	0.01	0.01
CO2	6.36	6.36	6.36	6.36	6.36	6.36	6.36	6.36
TDS	43,186.82	43,186.82	58,889.20	75,480.87	75,480.87	177.98	255.08	207.01
pH	7.60	7.60	7.72	7.82	7.82	5.81	5.96	5.86

Within Vessels - Pass1

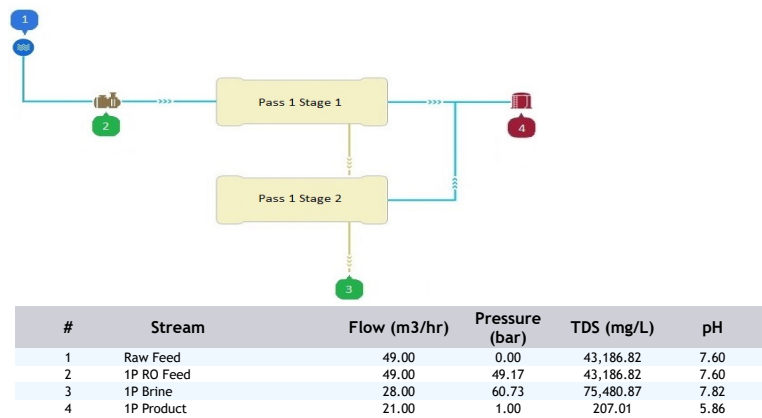
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.52	14.63	6.32	0.20	13.65	1.10	43,180.01	95.61
LG SW 400 R G2 UHP	2	7.65	0.43	12.17	5.62	0.18	11.71	1.09	46,088.94	120.80
LG SW 400 R G2 UHP	3	7.22	0.35	10.05	4.91	0.16	9.97	1.07	48,824.09	152.64
LG SW 400 R G2 UHP	4	6.87	0.29	8.25	4.24	0.15	8.43	1.06	51,339.18	192.64
LG SW 400 R G2 UHP	5	6.57	0.24	6.75	3.63	0.14	7.09	1.05	53,606.09	242.59
LG SW 400 R G2 UHP	6	6.34	0.19	5.51	3.07	0.13	5.94	1.04	55,614.32	304.57
LG SW 400 R G2 UHP	7	6.14	0.16	4.50	2.59	0.13	4.97	1.04	57,367.65	380.93
Stage 2										
LG SW 400 R G2 UHP	1	8.97	0.47	13.27	5.22	0.23	15.01	1.08	58,880.80	140.28
LG SW 400 R G2 UHP	2	8.50	0.39	10.96	4.55	0.21	12.97	1.07	62,115.38	176.53
LG SW 400 R G2 UHP	3	8.12	0.32	9.03	3.93	0.20	11.16	1.06	65,067.67	221.38
LG SW 400 R G2 UHP	4	7.80	0.26	7.44	3.37	0.19	9.58	1.05	67,719.13	276.41
LG SW 400 R G2 UHP	5	7.54	0.22	6.13	2.87	0.18	8.20	1.04	70,068.97	343.42
LG SW 400 R G2 UHP	6	7.32	0.18	5.06	2.44	0.17	7.01	1.04	72,129.43	424.32
LG SW 400 R G2 UHP	7	7.14	0.15	4.18	2.07	0.16	5.99	1.03	73,921.52	521.22

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	0.73	1.64
CaSO4	28.5 %	59.08 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.86	0.11

Warnings - Pass1

Disclaimer: LG Chem Design is intended to be used by persons having the requisite technical skill, at their own discretion and risk. When using LG Chem Design, it is the user's responsibility to make provisions against fouling, scaling and chemical attacks, to account for piping and valve pressure losses, feed pump suction pressure and permeate backpressure. LG Chem shall not be liable for any error or miscalculation in results obtained by using LG Chem Design. Because use conditions and applicable laws may differ from one location to another and may change with time,users are responsible for determining whether products are appropriate for their use.



Blank Page

Project name:
Customer:
Username:

Water type: Antofagasta 43K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 5
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 43,190.87 mg/L
Feed osmotic pressure: 30.4 bar
Concentrate osmotic pressure: 53.11 bar

Feed pressure: 50.06 bar (1P)

Specific energy: 4.88 kwh/m3

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.07 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 43,190.87 mg/L
Feed osmotic pressure: 30.4 bar
Concentrate osmotic pressure: 53.11 bar
Pump efficiency: 80 %

Temperature: 19 °C
Average NDP: 13.26 bar
Specific energy: 4.88 kwh/m3
Feed pressure: 50.06 bar
Permeate TDS: 270.22 mg/L
Fouling factor: 0.7

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.18	35.82	50.06	48.96	1.1	0	0	0	8.45	230.97
Stage 2	4	7	35.82	7.84	27.98	62.96	61.6	1.36	14	0	0	7.53	336.23

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	13,446.93	13,446.93	18,365.03	23,476.72	23,476.72	83.26	121.29	97.44
Potassium	473.76	473.76	646.84	826.66	826.66	3.46	5.03	4.05
Magnesium	1,462.12	1,462.12	1,999.48	2,558.91	2,558.91	1.96	2.87	2.30
Calcium	505.62	505.62	691.45	884.92	884.92	0.67	0.97	0.78
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	23,701.90	23,701.90	32,374.75	41,390.22	41,390.22	135.70	197.66	158.81
Sulfate	3,361.87	3,361.87	4,598.41	5,886.03	5,886.03	1.90	2.78	2.23
Nitrate	0.16	0.16	0.22	0.27	0.27	0.01	0.01	0.01
Carbonate	7.52	7.52	10.28	13.16	13.16	0.00	0.01	0.00
Bicarbonate	221.40	221.40	302.04	385.74	385.74	2.28	3.32	2.67
Boron	7.48	7.48	9.60	11.65	11.65	1.72	2.28	1.93
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.87	3.67	3.67	0.01	0.01	0.01
CO2	3.72	3.72	3.72	3.72	3.72	3.72	3.72	3.72
TDS	43,190.87	43,190.87	59,000.99	75,437.97	75,437.97	230.97	336.23	270.22
pH	7.60	7.60	7.67	7.73	7.73	6.07	6.22	6.13

Within Vessels - Pass1

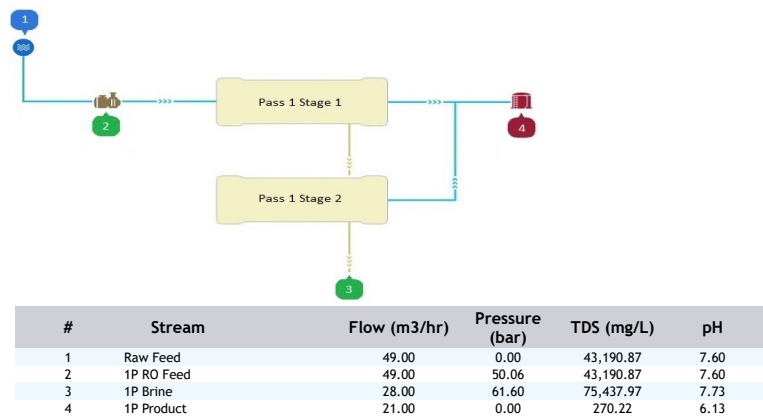
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.49	13.08	5.95	0.20	15.98	1.09	43,184.05	131.79
LG SW 400 R G2 UHP	2	7.68	0.42	11.20	5.42	0.18	14.06	1.08	45,908.63	161.73
LG SW 400 R G2 UHP	3	7.26	0.35	9.53	4.87	0.17	12.28	1.07	48,530.25	198.69
LG SW 400 R G2 UHP	4	6.91	0.30	8.06	4.33	0.15	10.66	1.06	51,006.76	244.12
LG SW 400 R G2 UHP	5	6.61	0.25	6.78	3.81	0.14	9.21	1.05	53,306.22	299.68
LG SW 400 R G2 UHP	6	6.36	0.21	5.70	3.33	0.13	7.92	1.05	55,408.20	367.24
LG SW 400 R G2 UHP	7	6.15	0.18	4.77	2.88	0.13	6.79	1.04	57,303.49	448.92
Stage 2										
LG SW 400 R G2 UHP	1	8.95	0.43	11.70	4.86	0.23	17.36	1.07	58,992.59	196.86
LG SW 400 R G2 UHP	2	8.52	0.37	9.95	4.34	0.22	15.35	1.06	61,993.60	240.56
LG SW 400 R G2 UHP	3	8.15	0.31	8.44	3.85	0.20	13.52	1.06	64,796.17	293.42
LG SW 400 R G2 UHP	4	7.84	0.27	7.14	3.39	0.19	11.88	1.05	67,377.70	356.86
LG SW 400 R G2 UHP	5	7.57	0.22	6.04	2.97	0.18	10.41	1.04	69,728.14	432.50
LG SW 400 R G2 UHP	6	7.35	0.19	5.12	2.59	0.17	9.11	1.04	71,847.33	522.01
LG SW 400 R G2 UHP	7	7.16	0.16	4.34	2.25	0.16	7.97	1.03	73,742.94	627.16

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	0.74	1.61
CaSO4	28.5 %	59.07 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.85	0.08

Warnings - Pass1

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Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 43K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 0
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 43,186.96 mg/L
Feed osmotic pressure: 30.92 bar
Concentrate osmotic pressure: 54.01 bar

Feed pressure: 47.9 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 43,186.96 mg/L
Feed osmotic pressure: 30.92 bar
Concentrate osmotic pressure: 54.01 bar
Pump efficiency: 80 %

Temperature: 24 °C
Average NDP: 9.23 bar
Specific energy: 4.53 kWh/m³
Feed pressure: 47.9 bar
Permeate TDS: 282.42 mg/L
Fouling factor: 1

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.07	35.93	47.9	46.82	1.08	0	1	0	8.81	243.61
Stage 2	4	7	35.93	7.95	27.98	60.82	59.48	1.34	14	1	0	8.05	346.17

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	13,445.45	13,445.45	18,302.57	23,470.59	23,470.59	87.39	124.26	101.34
Potassium	473.71	473.71	644.64	826.44	826.44	3.63	5.15	4.20
Magnesium	1,461.95	1,461.95	1,992.78	2,558.49	2,558.49	2.07	2.94	2.40
Calcium	505.56	505.56	689.14	884.77	884.77	0.71	1.01	0.82
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	23,704.52	23,704.52	32,271.79	41,388.97	41,388.97	142.80	203.05	165.60
Sulfate	3,362.24	3,362.24	4,584.06	5,886.50	5,886.50	2.00	2.85	2.32
Nitrate	0.16	0.16	0.22	0.27	0.27	0.01	0.01	0.01
Carbonate	2.37	2.37	3.23	4.15	4.15	0.00	0.00	0.00
Bicarbonate	221.40	221.40	300.72	384.89	384.89	3.27	4.64	3.79
Boron	7.48	7.48	9.57	11.65	11.65	1.73	2.25	1.93
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.86	3.67	3.67	0.01	0.01	0.01
CO2	5.87	5.87	5.87	5.87	5.87	5.87	5.87	5.87
TDS	43,186.96	43,186.96	58,801.58	75,420.40	75,420.40	243.61	346.17	282.42
pH	7.60	7.60	7.72	7.82	7.82	5.94	6.09	6.00

Within Vessels - Pass1

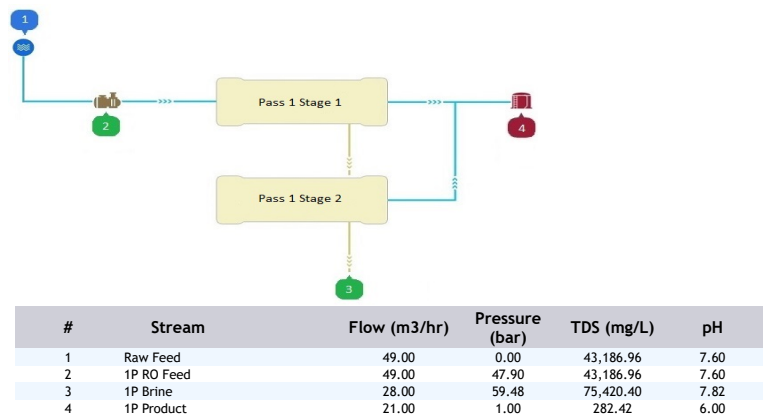
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.56	15.88	6.87	0.20	11.65	1.10	43,180.23	120.77
LG SW 400 R G2 UHP	2	7.61	0.45	12.69	5.89	0.18	9.64	1.08	46,354.98	159.01
LG SW 400 R G2 UHP	3	7.16	0.36	10.06	4.96	0.16	7.90	1.07	49,246.25	208.98
LG SW 400 R G2 UHP	4	6.80	0.28	7.95	4.12	0.15	6.44	1.06	51,807.40	273.66
LG SW 400 R G2 UHP	5	6.52	0.22	6.26	3.39	0.14	5.23	1.05	54,023.87	356.60
LG SW 400 R G2 UHP	6	6.30	0.17	4.94	2.77	0.13	4.23	1.04	55,906.72	461.79
LG SW 400 R G2 UHP	7	6.13	0.14	3.90	2.25	0.13	3.42	1.03	57,484.13	593.43
Stage 2										
LG SW 400 R G2 UHP	1	8.98	0.51	14.42	5.67	0.23	12.88	1.08	58,793.57	177.12
LG SW 400 R G2 UHP	2	8.47	0.40	11.46	4.78	0.21	10.77	1.07	62,316.09	231.86
LG SW 400 R G2 UHP	3	8.07	0.32	9.10	3.98	0.20	8.97	1.06	65,430.06	301.62
LG SW 400 R G2 UHP	4	7.75	0.26	7.23	3.30	0.19	7.45	1.05	68,130.85	389.49
LG SW 400 R G2 UHP	5	7.49	0.20	5.77	2.72	0.18	6.19	1.04	70,439.65	498.87
LG SW 400 R G2 UHP	6	7.29	0.16	4.62	2.24	0.17	5.14	1.03	72,393.01	633.15
LG SW 400 R G2 UHP	7	7.13	0.13	3.72	1.84	0.16	4.27	1.03	74,035.22	795.94

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	0.81	1.72
CaSO4	28.5 %	59.08 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.8	0.17

Warnings - Pass1

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Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 43K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 1
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 43,186.96 mg/L
Feed osmotic pressure: 30.92 bar
Concentrate osmotic pressure: 54 bar

Feed pressure: 48.39 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 43,186.96 mg/L
Feed osmotic pressure: 30.92 bar
Concentrate osmotic pressure: 54 bar
Pump efficiency: 80 %

Temperature: 24 °C
Average NDP: 9.76 bar
Specific energy: 4.57 kWh/m³
Feed pressure: 48.39 bar
Permeate TDS: 301.69 mg/L
Fouling factor: 0.93

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.1	35.9	48.39	47.3	1.09	0	1	0	8.83	259.64
Stage 2	4	7	35.9	7.92	27.98	61.3	59.96	1.34	14	1	0	8.01	371.22

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	13,445.45	13,445.45	18,316.97	23,466.22	23,466.22	93.15	133.27	108.27
Potassium	473.71	473.71	645.13	826.25	826.25	3.87	5.53	4.49
Magnesium	1,461.95	1,461.95	1,994.54	2,558.45	2,558.45	2.20	3.15	2.56
Calcium	505.56	505.56	689.74	884.76	884.76	0.75	1.08	0.88
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	23,704.52	23,704.52	32,297.45	41,381.96	41,381.96	152.21	217.77	176.92
Sulfate	3,362.24	3,362.24	4,588.16	5,886.59	5,886.59	2.13	3.06	2.48
Nitrate	0.16	0.16	0.22	0.27	0.27	0.01	0.01	0.01
Carbonate	2.37	2.37	3.23	4.15	4.15	0.00	0.00	0.00
Bicarbonate	221.40	221.40	300.91	384.71	384.71	3.48	4.97	4.04
Boron	7.48	7.48	9.55	11.58	11.58	1.82	2.37	2.03
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.86	3.67	3.67	0.01	0.01	0.01
CO2	5.87	5.87	5.87	5.87	5.87	5.87	5.87	5.87
TDS	43,186.96	43,186.96	58,848.76	75,408.63	75,408.63	259.64	371.22	301.69
pH	7.60	7.60	7.72	7.82	7.82	5.97	6.12	6.02

Within Vessels - Pass1

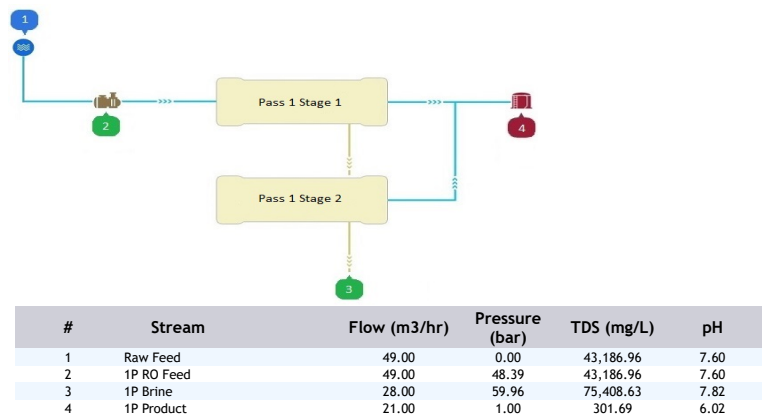
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.55	15.53	6.71	0.20	12.25	1.10	43,180.23	131.89
LG SW 400 R G2 UHP	2	7.62	0.44	12.56	5.82	0.18	10.24	1.08	46,279.08	171.60
LG SW 400 R G2 UHP	3	7.17	0.36	10.08	4.96	0.16	8.50	1.07	49,128.12	222.96
LG SW 400 R G2 UHP	4	6.82	0.28	8.05	4.17	0.15	7.00	1.06	51,679.99	288.82
LG SW 400 R G2 UHP	5	6.53	0.23	6.42	3.47	0.14	5.75	1.05	53,915.32	372.50
LG SW 400 R G2 UHP	6	6.31	0.18	5.11	2.86	0.13	4.70	1.04	55,838.30	477.72
LG SW 400 R G2 UHP	7	6.13	0.14	4.09	2.35	0.13	3.84	1.03	57,469.77	608.37
Stage 2										
LG SW 400 R G2 UHP	1	8.97	0.50	14.03	5.52	0.23	13.48	1.08	58,840.83	193.45
LG SW 400 R G2 UHP	2	8.48	0.40	11.28	4.70	0.21	11.38	1.07	62,266.32	250.26
LG SW 400 R G2 UHP	3	8.08	0.32	9.06	3.96	0.20	9.58	1.06	65,322.23	321.95
LG SW 400 R G2 UHP	4	7.76	0.26	7.28	3.31	0.19	8.04	1.05	67,999.82	411.42
LG SW 400 R G2 UHP	5	7.50	0.21	5.86	2.76	0.18	6.74	1.04	70,313.56	521.77
LG SW 400 R G2 UHP	6	7.30	0.17	4.74	2.29	0.17	5.65	1.03	72,292.86	656.21
LG SW 400 R G2 UHP	7	7.13	0.14	3.86	1.91	0.16	4.75	1.03	73,974.99	817.79

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	0.81	1.72
CaSO4	28.5 %	59.09 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.8	0.17

Warnings - Pass1

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Project name:
Customer:
Username:

Water type: Antofagasta 43K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 5
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 43,191.46 mg/L
Feed osmotic pressure: 30.92 bar
Concentrate osmotic pressure: 53.95 bar

Feed pressure: 48.89 bar (1P)

Specific energy: 4.79 kwh/m3

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.07 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 43,191.46 mg/L
Feed osmotic pressure: 30.92 bar
Concentrate osmotic pressure: 53.95 bar
Pump efficiency: 80 %

Temperature: 24 °C
Average NDP: 11.55 bar
Specific energy: 4.79 kwh/m3
Feed pressure: 48.89 bar
Permeate TDS: 393.4 mg/L
Fouling factor: 0.7

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.17	35.83	48.89	47.8	1.09	0	0	0	8.44	336.78
Stage 2	4	7	35.83	7.85	27.98	61.8	60.45	1.35	14	0	0	7.55	488.34

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	13,447.27	13,447.27	18,345.09	23,444.72	23,444.72	121.51	176.34	141.99
Potassium	473.77	473.77	646.05	825.33	825.33	5.04	7.31	5.89
Magnesium	1,462.15	1,462.15	1,998.51	2,558.28	2,558.28	2.88	4.17	3.36
Calcium	505.63	505.63	691.12	884.71	884.71	0.97	1.41	1.14
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	23,701.30	23,701.30	32,339.82	41,336.21	41,336.21	198.04	287.38	231.41
Sulfate	3,361.78	3,361.78	4,596.37	5,885.34	5,885.34	2.78	4.04	3.25
Nitrate	0.16	0.16	0.22	0.27	0.27	0.01	0.01	0.01
Carbonate	8.40	8.40	11.48	14.70	14.70	0.01	0.01	0.01
Bicarbonate	221.40	221.40	301.55	384.84	384.84	3.33	4.82	3.89
Boron	7.48	7.48	9.42	11.27	11.27	2.20	2.84	2.44
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.87	3.67	3.67	0.01	0.01	0.01
CO2	3.46	3.46	3.46	3.46	3.46	3.46	3.46	3.46
TDS	43,191.46	43,191.46	58,942.51	75,349.35	75,349.35	336.78	488.34	393.40
pH	7.60	7.60	7.67	7.73	7.73	6.22	6.37	6.28

Within Vessels - Pass1

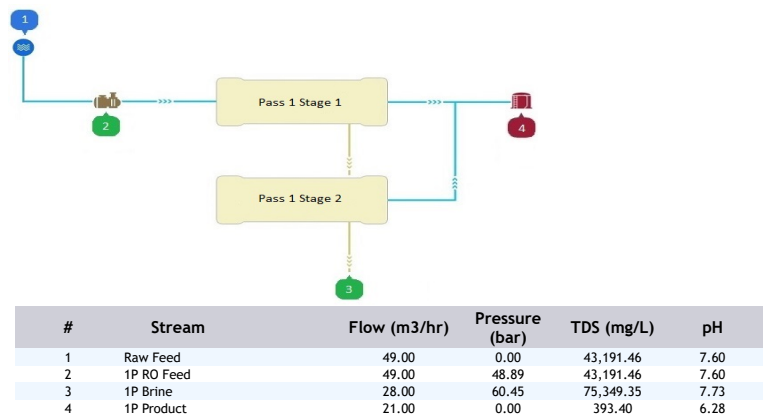
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.51	13.80	6.28	0.20	14.24	1.09	43,184.74	182.69
LG SW 400 R G2 UHP	2	7.65	0.43	11.54	5.60	0.18	12.26	1.08	46,065.62	229.84
LG SW 400 R G2 UHP	3	7.22	0.36	9.58	4.93	0.16	10.47	1.07	48,785.44	289.18
LG SW 400 R G2 UHP	4	6.87	0.29	7.91	4.28	0.15	8.89	1.06	51,298.08	363.39
LG SW 400 R G2 UHP	5	6.58	0.24	6.51	3.68	0.14	7.51	1.05	53,574.38	455.46
LG SW 400 R G2 UHP	6	6.33	0.20	5.35	3.14	0.13	6.32	1.04	55,602.34	568.66
LG SW 400 R G2 UHP	7	6.13	0.16	4.39	2.66	0.13	5.32	1.03	57,384.55	706.39
Stage 2										
LG SW 400 R G2 UHP	1	8.96	0.46	12.32	5.11	0.23	15.51	1.07	58,934.56	273.01
LG SW 400 R G2 UHP	2	8.50	0.38	10.24	4.48	0.21	13.45	1.06	62,093.24	341.57
LG SW 400 R G2 UHP	3	8.12	0.32	8.49	3.89	0.20	11.62	1.05	64,986.35	425.79
LG SW 400 R G2 UHP	4	7.80	0.26	7.04	3.35	0.19	10.01	1.04	67,596.19	528.29
LG SW 400 R G2 UHP	5	7.54	0.22	5.84	2.88	0.18	8.62	1.04	69,921.45	651.78
LG SW 400 R G2 UHP	6	7.33	0.18	4.86	2.46	0.17	7.41	1.03	71,973.35	798.95
LG SW 400 R G2 UHP	7	7.14	0.15	4.06	2.11	0.16	6.38	1.03	73,771.53	972.30

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	0.82	1.69
CaSO4	28.5 %	59.09 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.79	0.14

Warnings - Pass1

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Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 53K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 0
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 53,249.96 mg/L
Feed osmotic pressure: 37.49 bar
Concentrate osmotic pressure: 65.53 bar

Feed pressure: 60.63 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 53,249.96 mg/L
Feed osmotic pressure: 37.49 bar
Concentrate osmotic pressure: 65.53 bar
Pump efficiency: 80 %

Temperature: 19 °C
Average NDP: 12.67 bar
Specific energy: 5.65 kWh/m³
Feed pressure: 60.63 bar
Permeate TDS: 241.82 mg/L
Fouling factor: 1

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.85	35.14	60.63	59.56	1.06	0	1	0	9.34	198.05
Stage 2	4	7	35.14	7.16	27.98	75.56	74.24	1.32	16	1	0	7.25	326.45

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	16,578.86	16,578.86	23,086.77	28,969.17	28,969.17	71.10	117.26	86.83
Potassium	584.09	584.09	813.19	1,020.20	1,020.20	2.95	4.86	3.60
Magnesium	1,802.66	1,802.66	2,512.66	3,155.44	3,155.44	1.68	2.77	2.05
Calcium	623.38	623.38	868.91	1,091.20	1,091.20	0.57	0.95	0.70
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	29,229.56	29,229.56	40,707.01	51,083.09	51,083.09	116.17	191.61	141.89
Sulfate	4,145.79	4,145.79	5,779.55	7,259.02	7,259.02	1.63	2.69	1.99
Nitrate	0.20	0.20	0.28	0.34	0.34	0.01	0.01	0.01
Carbonate	2.83	2.83	3.94	4.95	4.95	0.00	0.00	0.00
Bicarbonate	273.00	273.00	379.58	475.67	475.67	2.66	4.38	3.24
Boron	7.48	7.48	9.93	11.98	11.98	1.27	1.91	1.49
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.93	3.67	3.67	0.01	0.01	0.01
CO2	7.68	7.68	7.68	7.68	7.68	7.68	7.68	7.68
TDS	53,249.96	53,249.96	74,164.77	93,074.75	93,074.75	198.05	326.45	241.82
pH	7.60	7.60	7.73	7.82	7.82	5.77	5.98	5.84

Within Vessels - Pass1

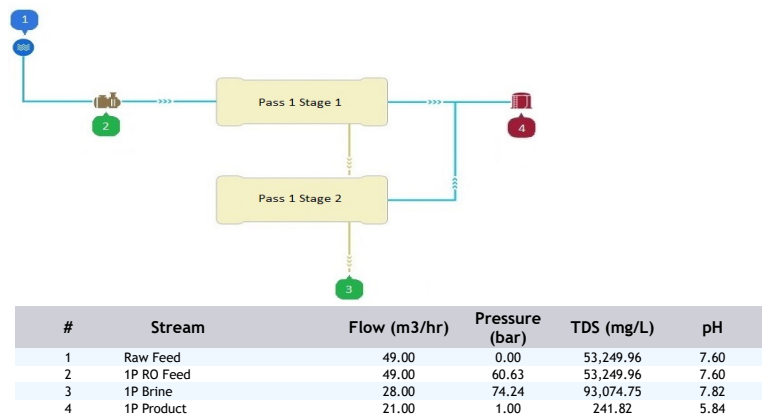
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.59	16.65	7.20	0.20	16.28	1.11	53,243.20	98.40
LG SW 400 R G2 UHP	2	7.58	0.47	13.34	6.21	0.18	13.73	1.10	57,365.25	129.51
LG SW 400 R G2 UHP	3	7.11	0.38	10.63	5.28	0.16	11.52	1.08	61,157.99	169.85
LG SW 400 R G2 UHP	4	6.73	0.30	8.45	4.43	0.15	9.62	1.07	64,557.57	221.55
LG SW 400 R G2 UHP	5	6.43	0.24	6.71	3.68	0.14	8.01	1.05	67,539.86	287.03
LG SW 400 R G2 UHP	6	6.20	0.19	5.35	3.05	0.13	6.66	1.04	70,112.50	368.97
LG SW 400 R G2 UHP	7	6.01	0.15	4.28	2.51	0.12	5.53	1.04	72,304.43	470.27
Stage 2										
LG SW 400 R G2 UHP	1	8.79	0.45	12.61	5.07	0.23	16.74	1.08	74,156.32	173.53
LG SW 400 R G2 UHP	2	8.34	0.36	10.11	4.28	0.21	14.48	1.07	78,105.63	224.08
LG SW 400 R G2 UHP	3	7.98	0.29	8.13	3.60	0.20	12.53	1.05	81,588.48	286.87
LG SW 400 R G2 UHP	4	7.70	0.23	6.57	3.01	0.18	10.86	1.05	84,620.48	363.71
LG SW 400 R G2 UHP	5	7.46	0.19	5.34	2.52	0.18	9.43	1.04	87,237.40	456.33
LG SW 400 R G2 UHP	6	7.28	0.15	4.37	2.12	0.17	8.21	1.03	89,485.07	566.34
LG SW 400 R G2 UHP	7	7.12	0.13	3.61	1.79	0.16	7.16	1.03	91,412.04	695.09

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	1	2.03
CaSO4	37.24 %	78.02 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.61	0.53

Warnings - Pass1

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Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 53K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 1
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 53,249.96 mg/L
Feed osmotic pressure: 37.49 bar
Concentrate osmotic pressure: 65.52 bar

Feed pressure: 61.36 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 53,249.96 mg/L
Feed osmotic pressure: 37.49 bar
Concentrate osmotic pressure: 65.52 bar
Pump efficiency: 80 %

Temperature: 19 °C
Average NDP: 13.44 bar
Specific energy: 5.71 kWh/m³
Feed pressure: 61.36 bar
Permeate TDS: 258.41 mg/L
Fouling factor: 0.93

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.89	35.11	61.36	60.3	1.07	0	1	0	9.37	211.16
Stage 2	4	7	35.11	7.13	27.98	76.3	74.97	1.32	16	1	0	7.22	350.41

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	16,578.86	16,578.86	23,106.01	28,965.00	28,965.00	75.81	125.88	92.80
Potassium	584.09	584.09	813.86	1,020.03	1,020.03	3.15	5.22	3.85
Magnesium	1,802.66	1,802.66	2,514.92	3,155.37	3,155.37	1.79	2.98	2.19
Calcium	623.38	623.38	869.69	1,091.17	1,091.17	0.61	1.02	0.75
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	29,229.56	29,229.56	40,741.19	51,076.36	51,076.36	123.87	205.69	151.64
Sulfate	4,145.79	4,145.79	5,784.81	7,259.00	7,259.00	1.74	2.89	2.13
Nitrate	0.20	0.20	0.28	0.34	0.34	0.01	0.01	0.01
Carbonate	2.83	2.83	3.95	4.95	4.95	0.00	0.00	0.00
Bicarbonate	273.00	273.00	379.85	475.50	475.50	2.83	4.70	3.47
Boron	7.48	7.48	9.91	11.92	11.92	1.34	2.02	1.57
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.93	3.67	3.67	0.01	0.01	0.01
CO2	7.68	7.68	7.68	7.68	7.68	7.68	7.68	7.68
TDS	53,249.96	53,249.96	74,227.41	93,063.34	93,063.34	211.16	350.41	258.41
pH	7.60	7.60	7.73	7.82	7.82	5.80	6.01	5.87

Within Vessels - Pass1

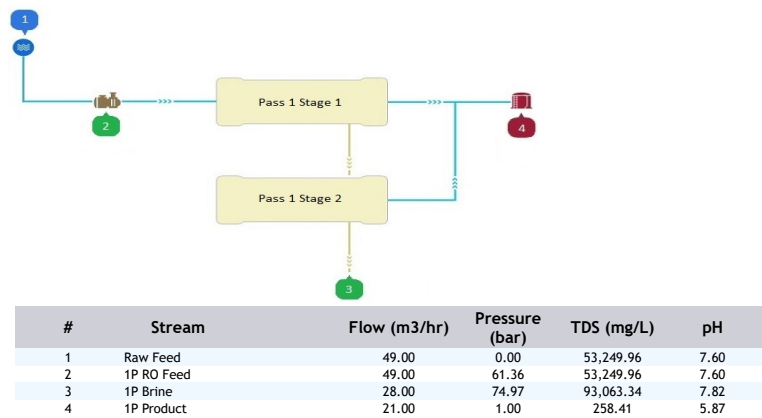
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.58	16.31	7.05	0.20	17.15	1.11	53,243.20	107.33
LG SW 400 R G2 UHP	2	7.59	0.47	13.22	6.15	0.18	14.61	1.09	57,275.17	139.72
LG SW 400 R G2 UHP	3	7.12	0.38	10.65	5.28	0.16	12.37	1.08	61,017.75	181.37
LG SW 400 R G2 UHP	4	6.75	0.30	8.55	4.47	0.15	10.43	1.07	64,406.35	234.32
LG SW 400 R G2 UHP	5	6.45	0.24	6.86	3.76	0.14	8.77	1.06	67,411.32	300.90
LG SW 400 R G2 UHP	6	6.20	0.19	5.52	3.14	0.13	7.37	1.05	70,032.51	383.64
LG SW 400 R G2 UHP	7	6.01	0.16	4.45	2.62	0.12	6.18	1.04	72,290.58	485.23
Stage 2										
LG SW 400 R G2 UHP	1	8.78	0.43	12.30	4.95	0.23	17.58	1.08	74,219.02	189.33
LG SW 400 R G2 UHP	2	8.34	0.35	9.97	4.22	0.21	15.33	1.06	78,073.05	242.00
LG SW 400 R G2 UHP	3	7.99	0.29	8.09	3.57	0.20	13.37	1.05	81,499.71	306.93
LG SW 400 R G2 UHP	4	7.71	0.23	6.59	3.02	0.18	11.68	1.05	84,508.56	385.85
LG SW 400 R G2 UHP	5	7.47	0.19	5.40	2.55	0.18	10.23	1.04	87,128.10	480.39
LG SW 400 R G2 UHP	6	7.28	0.16	4.46	2.16	0.17	8.97	1.03	89,397.20	592.04
LG SW 400 R G2 UHP	7	7.13	0.13	3.71	1.84	0.16	7.89	1.03	91,358.44	722.09

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	1	2.03
CaSO4	37.24 %	78.02 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.61	0.53

Warnings - Pass1

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Project name:
Customer:
Username:

Water type: Antofagasta 53k
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 5
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 53,256.51 mg/L
Feed osmotic pressure: 37.49 bar
Concentrate osmotic pressure: 65.48 bar

Feed pressure: 62.65 bar (1P)

Specific energy: 6 kwh/m3

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.07 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 53,256.51 mg/L
Feed osmotic pressure: 37.49 bar
Concentrate osmotic pressure: 65.48 bar
Pump efficiency: 80 %

Temperature: 19 °C
Average NDP: 16.02 bar
Specific energy: 6 kwh/m3
Feed pressure: 62.65 bar
Permeate TDS: 337.33 mg/L
Fouling factor: 0.7

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.96	35.04	62.65	61.57	1.07	0	0	0	8.94	274.15
Stage 2	4	7	35.04	7.06	27.98	77.57	76.24	1.33	16	0	0	6.79	462.17

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	16,581.51	16,581.51	23,146.27	28,947.70	28,947.70	98.97	166.95	121.82
Potassium	584.18	584.18	815.22	1,019.28	1,019.28	4.11	6.93	5.06
Magnesium	1,802.95	1,802.95	2,520.11	3,155.34	3,155.34	2.34	3.95	2.88
Calcium	623.48	623.48	871.49	1,091.17	1,091.17	0.79	1.34	0.98
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	29,224.86	29,224.86	40,800.46	51,032.36	51,032.36	161.29	272.11	198.53
Sulfate	4,145.12	4,145.12	5,795.17	7,257.27	7,257.27	2.26	3.82	2.78
Nitrate	0.20	0.20	0.28	0.34	0.34	0.01	0.01	0.01
Carbonate	11.61	11.61	16.23	20.33	20.33	0.01	0.01	0.01
Bicarbonate	273.00	273.00	380.65	475.60	475.60	2.71	4.57	3.34
Boron	7.48	7.48	9.80	11.65	11.65	1.65	2.46	1.92
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.93	3.67	3.67	0.01	0.01	0.01
CO2	4.19	4.19	4.19	4.19	4.19	4.19	4.19	4.19
TDS	53,256.51	53,256.51	74,358.62	93,014.73	93,014.73	274.15	462.17	337.33
pH	7.60	7.60	7.67	7.72	7.72	6.09	6.30	6.17

Within Vessels - Pass1

	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.54	14.57	6.63	0.20	20.06	1.10	53,249.75	147.33
LG SW 400 R G2 UHP	2	7.62	0.45	12.18	5.94	0.18	17.53	1.09	57,020.13	185.89
LG SW 400 R G2 UHP	3	7.17	0.38	10.12	5.25	0.16	15.23	1.08	60,607.74	234.33
LG SW 400 R G2 UHP	4	6.80	0.31	8.38	4.58	0.15	13.17	1.06	63,949.81	294.68
LG SW 400 R G2 UHP	5	6.48	0.26	6.92	3.96	0.14	11.35	1.06	67,005.41	369.11
LG SW 400 R G2 UHP	6	6.23	0.21	5.71	3.41	0.13	9.77	1.05	69,755.53	460.05
LG SW 400 R G2 UHP	7	6.02	0.18	4.72	2.92	0.12	8.39	1.04	72,199.37	569.91
Stage 2										
LG SW 400 R G2 UHP	1	8.76	0.40	10.89	4.62	0.23	20.41	1.07	74,350.22	264.64
LG SW 400 R G2 UHP	2	8.36	0.34	9.08	4.04	0.21	18.18	1.06	77,937.14	328.42
LG SW 400 R G2 UHP	3	8.02	0.28	7.58	3.51	0.20	16.19	1.05	81,204.17	405.34
LG SW 400 R G2 UHP	4	7.74	0.24	6.34	3.05	0.19	14.42	1.04	84,146.80	497.05
LG SW 400 R G2 UHP	5	7.50	0.20	5.32	2.64	0.18	12.87	1.04	86,775.34	605.06
LG SW 400 R G2 UHP	6	7.30	0.17	4.49	2.29	0.17	11.51	1.03	89,110.14	730.69
LG SW 400 R G2 UHP	7	7.14	0.14	3.81	1.98	0.16	10.31	1.03	91,177.29	875.02

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	1.01	2
CaSO4	37.23 %	78.02 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.6	0.5

Warnings - Pass1

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